Bonded Mirror Silver

Bonded metallic base with UV-stable clear coat · RAL 9006

RAL 9006 · White aluminium

Product	Code • series	Colour reference	Document
Bonded Mirror Silver 邦定镜面银	BL-M103 BL-M Bonded Metallics	RAL 9006 · White aluminium	TDS BL-M103 Rev. 2026-10

P1000 · tested to AAMA 2603 / QUALICOAT I test items on request — no product certificate held

1. Description

A chrome-like finish without electroplating: no hexavalent chromium, no plating waste water, no rack marks. It is a two-coat system — the base alone is a metallic grey and all the gloss comes from the clear. Selling it as a single coat is the most common way this product fails.

2. Typical properties

Property	Value	Method
Chemistry	Bonded metallic base with UV-stable clear coat	—
Colour reference	RAL 9006 White aluminium — approval on a sprayed panel against a physical chip	RAL
Finish and gloss	90–100 @60°	ASTM D523
Film build	60–80 µm + clear	ISO 2808
Density	1.6–1.8 g/cm³	ISO 8130-2/3
Coverage	Base 8.4 m²/kg theoretical; the base + clear pair covers 5.2 m²/kg theoretical, 4.9 m²/kg at 95 % utilisation (with reclaim)	calculated
Cure schedule	180–200 °C × 10–15 min (object temperature)	object temp., not oven set point
Particle size D50	30–45 µm	ISO 8130-1 / -13
Stoving loss	≤ 1–2 %	ISO 8130-7
Storage stability	1 month at 35–40 °C	ISO 8130-8

3. Performance data

Test	Method	Typical result
1. Adhesion, cross-cut	ISO 2409 / ASTM D3359	0–1
2. Impact, direct and reverse	ASTM D2794	50 kg·cm, no cracking
3. Pencil hardness	ASTM D3363	H base, 2H with clear
4. Gloss 60°	ASTM D523	90–100 (with clear coat)
5. Colour difference	ASTM D2244	ΔE ≤ 1.5 same batch
6. Neutral salt spray	ASTM B117 / ISO 9227	500 h
7. QUV-A	ASTM G154	500 h with the UV-stable clear
8. Abrasion	ASTM D968	Per clear-coat grade — specify if the part is handled daily

Measured to the method listed, on production batches or the third-party report held for this system. Not a certificate: acceptance values are fixed in the trial report and the purchase specification.

4. Application and cure

01 Pre-treatment

As BL-M102: conversion coating on aluminium, phosphate on steel, DI rinse, dry. A mirror finish shows every pre-treatment defect — this is the one product where the rinse water quality is visible.

02 Base coat

60–80 µm, corona 60–90 kV, 150–250 mm. Keep the DFT window tight: film build variation shows as a grey-to-silver gradient across a large panel.

03 Clear coat

40–60 µm over a gelled base — powder-on-powder. The gel window is fixed on the trial: too little gel and the clear sinks, too much and inter-coat adhesion drops.

04 Cure

Base 180–200 °C × 10–15 min; clear 180–200 °C × 10–15 min. Measure object temperature on both passes.

05 Handling

Do not touch the base before the clear is applied and cured. A mark under the clear cannot be polished out — it is a reject.

5. Packaging, storage and ordering

1. Standard pack: 20 kg net, polyethylene-lined carton. Other pack sizes on request.
2. Store below 30 °C, dry, out of direct sunlight, in the unopened pack. Keep away from ovens and hot lines.
3. Shelf life: 12 months from the production date in the original pack. Storage stability is verified at 35–40 °C for one month (ISO 8130-8).
4. A batch test report accompanies every delivery; a retained sample is kept for 12 months.
5. Sample in 3 days; production lead time confirmed per colour and pack size at order.
6. Colour match ΔE ≤ 1.5 within a batch. Cross-batch tolerance is agreed in the purchase specification.
7. Ask for: SDS · third-party test report · batch test report · physical colour chip.

6. Health and safety

1. Not classified as dangerous goods for transport. The SDS is available for every product — ask for it with the TDS.
2. The powder is a combustible dust. Earth the workpiece, the gun and the booth; no open flame or hot work in the booth; extract booth air; avoid dust clouds when emptying or cleaning.
3. Wear a P2 particulate mask, goggles and gloves when handling dry powder or cleaning the booth.
4. Cure volatiles: extract the oven and let parts cool before handling. Do not enter the oven zone without ventilation.

7. Also in this range

Code	Product	Colour	Film build	Cure
BL-M102	Bonded Gold Sand	RAL 1036	70–90 µm	200 °C × 10 min

8. Document control

Sheet	Revision	Issued	Supersedes	Languages
BL-M103 TDS	2026-10	2026-10	None — first issue	English + 中文 (4 pp.)

9. Limits of this document

What this product is not for:

— Never quote it as a single-coat product. Without the clear it greys out and every fingerprint stays.

— Exterior use only with the UV-stable clear, and the reference class is still P1000.

— If the part is a light reflector, specify gloss and orange-peel limits separately: a mirror look and an optical surface are not the same requirement.

1. Values in this sheet are typical for the product and were measured to the methods listed. They are not a certificate and not a warranty of fitness for a particular purpose. Acceptance values are fixed in the trial report and the purchase specification.
2. Colour: RAL references only. RAL does not define sRGB values, so the swatch in this sheet is a digital approximation — approval is made on a sprayed panel against a physical RAL chip.
3. No AAMA 2603 / 2604 / 2605, QUALICOAT, GSB or UL product certificate is held. We test to those methods and can supply third-party reports and in-house batch reports; if a tender requires the certificate itself, we will say plainly that we do not have it.
4. The 4,000 h salt spray and 10-year South Florida results shown elsewhere on our site belong to the fluorocarbon (PVDF) system, not to this powder product.

Issued by Guangdong Baolilai Industrial Co., Ltd. · jerry888.xu@gmail.com · +86 188 2310 0890
Ask for: SDS · third-party test report · batch report · physical colour chip · sample in 3 days

邦定镜面银

邦定金属底粉 + 耐候透明罩光 · RAL 9006

RAL 9006 · 白铝色

产品	编号 • 系列	参照颜色	文件
Bonded Mirror Silver 邦定镜面银	BL-M103 BL-M 邦定金属	RAL 9006 · 白铝色	TDS BL-M103 Rev. 2026-10

P1000 · 可按 AAMA 2603 / QUALICOAT I 测试项目送检（可索取报告）——不持有产品认证证书

1. 产品描述

不用电镀的类铬镜面：无六价铬、无电镀废水、无挂具印。它是双涂层体系——只有底粉就是一层金属灰，光泽全部来自罩光。把它当单涂层卖，是这款产品最常见的失败原因。

2. 典型性能

项目	数值	方法
体系	邦定金属底粉 + 耐候透明罩光	—
参照颜色	RAL 9006 白铝色——以喷板对照实物色卡确认	RAL
表面与光泽	90–100（60°）	ASTM D523
膜厚	60–80 µm + clear	ISO 2808
比重	1.6–1.8 g/cm³	ISO 8130-2/3
覆盖率	底粉理论 8.4 m²/kg；底粉+罩光两层理论 5.2 m²/kg，按 95% 利用率（含回收）4.9 m²/kg	计算值
固化条件	180–200 °C × 10–15 min（工件温度）	测工件温度，非炉膛设定
粒径 D50	30–45 µm	ISO 8130-1 / -13
烘烤失重	≤ 1–2 %	ISO 8130-7
贮存稳定性	35–40 °C 下 1 个月	ISO 8130-8

3. 性能数据

检测项目	方法	典型结果
1. 划格附着力	ISO 2409 / ASTM D3359	0–1
2. 正反冲击	ASTM D2794	50 kg·cm 不裂
3. 铅笔硬度	ASTM D3363	底粉 H，加罩光 2H
4. 光泽 60°	ASTM D523	90–100（含罩光）
5. 色差	ASTM D2244	同批 ΔE ≤ 1.5
6. 中性盐雾	ASTM B117 / ISO 9227	500 h
7. QUV-A 加速老化	ASTM G154	配耐候罩光 500 h
8. 耐磨	ASTM D968	按罩光等级——日常频繁接触件请指定

按所列方法在量产批次或本体系所持第三方报告上测定。不是证书：验收值以试粉报告与采购规格为准。

4. 施工与固化

01　前处理

同 BL-M102：铝件转化膜、钢件磷化、纯水洗、干燥。镜面会把前处理缺陷全部显出来——这是唯一一款“水洗水质看得见”的产品。

02　底粉

60–80 µm，电晕 60–90 kV，枪距 150–250 mm。膜厚窗口要收紧：膜厚波动在大板上会表现为“灰到银”的渐变。

03　罩光

在已胶化的底粉上喷 40–60 µm——粉上粉。胶化窗口在试粉时确定：胶化不足罩光会陷，过度则层间附着下降。

04　固化

底粉 180–200 °C × 10–15 min；罩光 180–200 °C × 10–15 min。两道均测工件温度。

05 搬运

未上罩光并固化前不要触碰底粉。罩光下的印记抛不掉——只能报废。

5. 包装、贮存与订货

1. 标准包装：20 kg 净重，纸箱内衬聚乙烯袋。其他包装规格可订。
2. 30 °C 以下、干燥、避免阳光直射，保持原包装未开封。远离烘道与高温工位。
3. 保质期：原包装自生产日起 12 个月。贮存稳定性按 ISO 8130-8 于 35–40 °C 验证一个月。
4. 每批附批次检测报告；留样保存 12 个月。
5. 3 天出样；生产交期按颜色与包装规格在订单时确认。
6. 同批色差 ΔE ≤ 1.5；跨批公差在采购规格中约定。
7. 可索取：SDS · 第三方检测报告 · 批次检测报告 · 实物色卡。

6. 健康与安全

1. 不属于运输危险品。每个产品均提供 SDS（安全数据表），可与 TDS 一并索取。
2. 粉末为可燃性粉尘。工件、喷枪与喷房必须接地；喷房内禁止明火与动火作业；喷房须抽风；倒粉与清房时避免扬尘。
3. 处理干粉或清理喷房时，佩戴 P2 防尘口罩、护目镜与手套。
4. 固化挥发物：烘道须抽风；工件冷却后再搬运。无通风不得进入烘道区域。

7. 同系列其他产品

编号	产品	颜色	膜厚	固化
BL-M102	邦定金砂	RAL 1036	70–90 µm	200 °C × 10 min

8. 文件控制

文件	版本	签发	替代版本	语言
BL-M103 TDS	2026-10	2026-10	无——首版	英文 + 中文（共 4 页）

9. 本文件的边界

这款产品不适合用在：

— 绝不能按单涂层报价。没有罩光，它会发灰，而且每个指纹都留得住。

— 户外仅可配耐候罩光使用，参照等级仍为 P1000。

— 若工件是灯具反射罩，请单独规定光泽与桔皮限度：“看起来像镜面”和“光学面”不是一个要求。

1. 本表数值为该产品的典型值，按所列方法测定。它们不是证书，也不构成对特定用途适用性的保证。验收值以试粉报告与采购规格为准。
2. 颜色：仅作 RAL 参照。RAL 未定义 sRGB 数值，本表色块为数字近似——以喷涂样板对照实物 RAL 色卡确认。
3. 我们不持有 AAMA 2603 / 2604 / 2605、QUALICOAT、GSB 或 UL 产品认证证书。我们按这些方法检测，可提供第三方报告与厂内批次报告；若招标硬性要求证书本身，我们会直接说明没有。
4. 我们网站上 4000 小时盐雾与 10 年南佛罗里达曝晒的结果属于氟碳（PVDF）体系，不属于本粉末产品。

签发：广东宝利莱工业有限公司 · jerry888.xu@gmail.com · +86 188 2310 0890
可索取：SDS · 第三方检测报告 · 批次报告 · 实物色卡 · 3 天出样